

Heart Failure: A Comprehensive Disease Characterization Report

Disease: Heart Failure | **MONDO ID:** MONDO:0005252 | **Category:** Complex **ICD-10:** I50 | **ICD-11:** BD10-BD1Z | **MeSH:** D006333 | **Date:** 2026-08-23

Summary

Heart failure (HF) is a complex, highly prevalent clinical syndrome — not a single disease — defined by the 2021 Universal Definition as symptoms and/or signs caused by a structural and/or functional cardiac abnormality, corroborated by elevated natriuretic peptides and/or objective evidence of pulmonary or systemic congestion. It affects **more than 64 million people worldwide**, carries a **lifetime risk of approximately 1 in 4**, and has five-year mortality rates that rival solid-organ malignancies. HF is classified by left ventricular ejection fraction (LVEF) into HFrEF ($\leq 40\%$), HFmrEF (41–49%), HFpEF ($\geq 50\%$), and HFimpEF (improved), and staged A (at-risk) through D (advanced) — with roughly two-thirds of US adults sitting in the pre-HF stages A/B, defining an enormous population reservoir for prevention.

Diverse etiologic insults — coronary ischemia, hypertension, genetic cardiomyopathy (notably *TTN*-truncating and *LMNA* variants), infection (Chagas disease from *Trypanosoma cruzi*), and cardiotoxic/metabolic exposures (obesity, diabetes, air pollution, trace-element dysregulation) — converge on a **shared pathophysiology**: neurohormonal activation (RAAS and sympathetic nervous system), mitochondrial energetic failure and oxidative stress, and inflammation-driven fibrotic remodeling. This causal chain produces progressive pump dysfunction and congestion. The syndrome is diagnosed with natriuretic peptides (BNP/NT-proBNP) plus imaging (echocardiography), and increasingly refined by multimarker and multi-omic profiling that reveals etiology-specific biology.

Therapeutically, HFrEF is now highly treatable: four foundational **guideline-directed medical therapy (GDMT)** pillars — RAAS inhibition/ARNI, beta-blockers, mineralocorticoid receptor antagonists (MRA), and SGLT2 inhibitors — substantially reduce mortality and hospitalization, with SGLT2i extending benefit across the EF spectrum. Advanced disease is managed with

device therapy (CRT/ICD), left ventricular assist devices (LVAD), and heart transplantation. Prevention is tiered — primary risk-factor/lifestyle control being the highest-yield strategy given the vast pre-HF population. HF is strongly conserved across mammals (naturally occurring canine and feline cardiomyopathy) and well-modeled in rodents (transverse aortic constriction, *Ttn*-truncation knock-in), zebrafish (HFpEF), and human iPSC-cardiomyocytes.

Key Findings

Finding 1 — Heart failure is a clinical syndrome classified by ejection fraction and staged by risk

The **2021 Universal Definition and Classification of Heart Failure** (jointly issued by the Heart Failure Society of America, the Heart Failure Association of the ESC, and the Japanese Heart Failure Society) established that "*HF is a clinical syndrome with symptoms and or signs caused by a structural and/or functional cardiac abnormality and corroborated by elevated natriuretic peptide levels and or objective evidence of pulmonary or systemic congestion*" [PMID: 33663906](#). The EF-based classification defines HFrEF (LVEF $\leq 40\%$), HFmrEF (41–49%), HFpEF ($\geq 50\%$), and HFimpEF (baseline $\leq 40\%$, ≥ 10 -point rise to $>40\%$), and stages the disease A (at-risk), B (pre-HF), C (symptomatic), and D (advanced).

A single-center registry of 8,471 patients applying this definition confirmed the real-world distribution: "*The most frequent type of HF was HFrEF ($n = 4947$; 58.4%), followed by HFpEF ($n = 1138$; 28.2%) and HFmrEF ($n = 2386$; 13.4%)*" [PMID: 38493451](#), with all-cause death highest in HFrEF (42.7%). **Ontology:** MONDO:0005252; ICD-10 I50; MeSH D006333.

HF subtype	LVEF	Registry frequency	All-cause mortality
HFrEF	$\leq 40\%$	58.4%	42.7% (highest)
HFpEF	$\geq 50\%$	28.2%	intermediate
HFmrEF	41–49%	13.4%	lowest

Finding 2 — Lifetime risk is ~1 in 4, with a vast pre-HF reservoir

The lifetime risk of developing HF is approximately **1 in 4**: "*The lifetime risk of developing HF stands at 1 in 4*" PMID: 41670570. Critically, "*roughly two-thirds of adults in the United States [are] classified as either stage A (at risk) or stage B (pre-HF),*" and "*many HF cases arise from nonatherosclerotic causes, such as obesity, hypertension, cancer therapies, and the cardio-kidney-metabolic syndrome.*" This enormous pre-HF population is the central target for primary prevention, and the rising incidence of HFpEF across contemporary cohorts underscores the shift toward metabolic and non-ischemic drivers.

Finding 3 — Genetic dilated cardiomyopathy: *TTN* and *LMNA* dominate causal genetics

In monogenic/familial dilated cardiomyopathy (DCM), a leading cause of HFrEF, disease-causing variants are found in ~25–46% of patients. In a 280-patient Polish DCM cohort, disease-causing variants were found in 46%, and "*Variants in titin (*TTN*) and lamin A/C (*LMNA*) genes were the most frequent (18% and 8% of the study cohort, respectively)*" PMID: 40792443. A multinational cohort of 2,088 DCM patients found: "*Of the 2,088 patients 514 (24.6%) obtained a molecular diagnosis; 534 LP/P variants were observed across 45 genes*" PMID: 37795486, spanning sarcomere, nuclear lamina, desmosome, cytoskeleton, and mitochondrial proteins. *LMNA*-related DCM carries genotype-specific high risk: "*the risks of severe DCM and the composite end point were 2.4- and 3-fold higher, respectively, for LMNA-related DCM*" [PMID: 40792443]. A review confirmed "*The most common genes related to dilated cardiomyopathy include *TTN*, *LMNA*, *MYH7**" PMID: 40155570.

HGNC causal genes: *TTN*, *LMNA*, *MYH7*, *MYBPC3*, *TNNT2*, *DSP*, *FLNC*, *BAG3*, *SCN5A*, *RBM20*, *DMD*. Population screening data (ARIC + UK Biobank) show ~0.7% of individuals carry an actionable cardiomyopathy variant, conferring increased risk of HF (HR 1.7), atrial fibrillation (HR 2.9), and all-cause mortality (HR 1.5) PMID: 35544052.

Finding 4 — Four GDMT pillars reduce mortality and hospitalization in HFrEF

GDMT for HFrEF comprises four foundational pharmacologic pillars: RAAS inhibition or angiotensin receptor-neprilysin inhibition (ARNI, sacubitril/valsartan), evidence-based beta-blockers, mineralocorticoid receptor antagonists (MRA), and SGLT2 inhibitors. Robust evidence supports "*early and combined use of the four foundational GDMT classes, with substantial reductions in mortality and heart failure hospitalization*" PMID: 42016211, advocating rapid "four drugs in 4 weeks" initiation. SGLT2i (DAPA-HF, EMPEROR-Reduced, EMPEROR-Preserved,

DELIVER) reduce HF hospitalization across the EF spectrum by promoting "*a shift in substrate utilization toward fatty acids and ketone bodies, improving mitochondrial efficiency*" and reducing oxidative stress and inflammation [PMID: 41913872](#). Emerging additional pillars include vericiguat (soluble guanylate cyclase stimulator) and digitoxin [PMID: 42329367](#). In HFmrEF/HFpEF, combined SGLT2i + finerenone reduced CV death/worsening HF by 31% (HR 0.69; 95% CI 0.59–0.81) [PMID: 41052644](#).

NCIT interventions: sacubitril/valsartan, dapagliflozin, empagliflozin, spironolactone/eplerenone, carvedilol/metoprolol/bisoprolol.

Finding 5 — Pathophysiology: neurohormonal activation and mitochondrial dysfunction drive remodeling

HF congestion is "*driven by neurohormonal dysregulation involving the renin-angiotensin-aldosterone system and sympathetic nervous system*," producing dyspnea, peripheral edema, and fatigue [PMID: 40801414](#). Mitochondrial dysfunction is a central driver: impaired oxidative phosphorylation, ROS imbalance, disrupted calcium handling, and defective mitophagy. Post-MI cardiomyocytes exhibit "*a reduction in total mitochondrial membrane potential (MMP) and an increase in reactive oxygen species levels*" [PMID: 42411500](#). Impaired inter-organelle mitochondrial communication produces "*dysregulated energy metabolism, oxidative stress, lipotoxicity, and impaired cardiomyocyte function*," with phenotype-specific differences between HFpEF and HFrEF [PMID: 42209899](#). The lysosomal/autophagy regulator TFEB links afterload-induced energy demand to hypertrophy: TFEB knockout hearts under pressure overload "*manifested an amplified hypertrophic response, leading rapidly to HF*" [PMID: 42299666](#).

GO terms: GO:0006119 (oxidative phosphorylation), GO:0006979 (response to oxidative stress), GO:0006954 (inflammatory response), GO:0006936 (muscle contraction). **CL terms:** CL:0000746 (cardiac muscle cell), CL:0000057 (fibroblast).

Finding 6 — Core phenotypes cluster into symptom groups that impair quality of life

HF symptoms cluster into five groups — emotional, digestive, ischemic (dizziness, chest pain, palpitations, fatigue), dyspnea (orthopnea, paroxysmal nocturnal dyspnea, sleep difficulty), and congestion (cough, shortness of breath, edema) [PMID: 39450907](#). Symptom burden strongly impairs health-related quality of life (measured by KCCQ and the Minnesota Living with Heart Failure Questionnaire). End-stage HF is "*often complicated with symptoms such as*

dyspnea, fatigue, pain, and nausea, which can worsen the quality of life" [PMID: 41550625](#). NYHA classes I–IV grade severity; symptoms are adult/late-onset and progressive with episodic decompensation.

HPO terms: HP:0001635 (congestive heart failure), HP:0002094 (dyspnea), HP:0012651 (orthopnea), HP:0012764 (paroxysmal nocturnal dyspnea), HP:0012378 (fatigue), HP:0100598 (peripheral edema), HP:0001962 (palpitations).

Finding 7 — Natriuretic peptides are diagnostic and prognostic cornerstones

BNP and NT-proBNP "*remain the cornerstones of biomarker-guided management, yet their interpretation is influenced by age, renal function, obesity, and atrial fibrillation*" [PMID: 42336497](#). In 116,466 diabetics without known HF, NT-proBNP >300 pg/mL predicted incident HF/death with HR 4.48 (95% CI 3.11–6.47) in T1D and 3.58 (3.39–3.78) in T2D versus <125 pg/mL [PMID: 41166576](#). Complementary biomarkers with incremental prognostic value include high-sensitivity troponin (myocardial injury), hsCRP (inflammation), galectin-3 and sST2 (fibrosis/remodeling), GDF-15, MPO, MR-proADM, MR-proANP [PMID: 41883058](#), copeptin [PMID: 41467274](#), and circulating non-coding RNAs.

LOINC: NT-proBNP 33762-6; BNP 30934-4; hs-troponin T 67151-1. Diagnostic screening thresholds: BNP ≥ 50 or NT-proBNP ≥ 125 pg/mL.

Finding 8 — Major modifiable risk factors dominate a multifactorial etiology

"*Its prevalence rises significantly with age. In addition, age-related conditions, such as hypertension, coronary artery disease, obesity, and diabetes mellitus, contribute to an elevated risk of heart failure*" [PMID: 42446811](#). Obesity promotes HF "*through complex metabolic, haemodynamic, inflammatory, and endothelial mechanisms,*" and bariatric surgery reduces these risk factors [PMID: 42591586](#). Environmental exposures — "*air pollution, noise, heat, chemical contamination, and light pollution*" — increase HF incidence via "*oxidative stress, inflammation, endothelial dysfunction, and circadian disruption,*" even below regulatory thresholds [PMID: 42095252](#). HF is thus multifactorial/polygenic: modifiable factors (hypertension, CAD, diabetes, obesity, smoking, pollution) dominate; age and family history are non-modifiable.

Finding 9 — Global burden rivals malignancy; iron deficiency is prevalent and treatable

"Heart failure (HF) affects over 64 million people worldwide, with five-year mortality rates rivaling solid-organ malignancies" PMID: 42606681. Iron deficiency "affected approximately 30-58% of patients with HF," and in HFrEF, "intravenous ferric carboxymaltose improved Patient Global Assessment (odds ratio [OR] 2.51; 95% confidence interval [CI] 1.75-3.61) and reduced HF hospitalization (hazard ratio [HR] 0.39; 95% CI 0.19-0.82)." Prognostic factors include EF class, baseline cardiac/renal function, female sex, mechanical ventilation, and sinus rhythm. LVAD destination-therapy on-device survival is 82.3%/73.3%/49.8% at 1/3/5 years PMID: 42629526.

Finding 10 — Advanced therapies: devices, LVAD, and transplantation with reverse remodeling

Device therapy includes cardiac resynchronization therapy (CRT) and implantable cardioverter-defibrillators (ICD); CRT upgrade "was associated with lower all-cause mortality, heart transplant, or assist device implant" PMID: 40533423. LVADs, used as bridge-to-transplant and destination therapy, "reverse the molecular transformations that take place in the cardiomyocytes... eventually leading to partial or complete recovery in a subset of patients" PMID: 40387335. Heart transplantation is definitive for Stage D disease. Cardiac rehabilitation improves 6MWT distance, METs, VO₂, LVEF, and anxiety/depression PMID: 40955678. NCIT: cardiac resynchronization therapy, implantable cardioverter-defibrillator, ventricular assist device, heart transplantation, cardiac rehabilitation.

Finding 11 — Robust model organisms recapitulate HF pathology

Transverse aortic constriction (TAC) in mice "induced progressive systolic dysfunction, fibrosis, and reduced microvascular density" — the standard pressure-overload model PMID: 33464950. A *Ttn*-truncation knock-in mouse recapitulates human DCM: "a pattern of DCM can be induced by TAC-mediated pressure overload in a *TTN*-truncated mouse model" PMID: 26504781. A zebrafish ion-imbalance model "recapitulates key characteristics of incipient heart failure with preserved ejection fraction" PMID: 36536484. In vitro models include H9c2 cells, neonatal rat/mouse ventricular cardiomyocytes, and human iPSC-cardiomyocytes.

Taxa: mouse (NCBI:txid10090), rat (10116), zebrafish (7955). **Resources:** MGI, RGD, ZFIN.

Finding 12 — Naturally occurring cross-species HF/DCM: the Doberman comparative model

"Dilated cardiomyopathy (DCM) is a naturally occurring heart failure condition in humans and dogs, notably characterized by a reduced contractility and ejection fraction" PMID: 37505469. Canine DCM shares conserved mechanisms: *"IDCM is associated with a marked impairment of mitochondrial production of ATP, arising from decreased activity of the mitochondrial electron transport system"* PMID: 1338376, and an autoimmune driver — β 1-adrenergic receptor autoantibodies (β 1-AAB) that are *"highly present in patients with dilated cardiomyopathy (DCM) and are increasingly accepted as disease drivers"* PMID: 32212256. **Species:** dog (NCBI:txid9615), cat (9685). **Breeds:** Doberman Pinscher, Boxer, Great Dane (canine DCM); Maine Coon, Ragdoll (feline HCM). **Resource:** OMIA.

Finding 13 — Infectious/inflammatory etiologies: Chagas cardiomyopathy has the worst prognosis

Chagas disease, caused by *Trypanosoma cruzi* (NCBI:txid5693), *"is a common cause of heart failure (HF) in Latin America and has recently been declared endemic in the United States"* PMID: 41847775. In a pooled analysis of 23,647 HFrEF patients, *"Chagasic HF had the highest incidence rates of all clinical outcomes compared with other etiologies,"* with adjusted HRs versus ischemic HF of 1.65 (composite), 1.86 (CV death), and 2.16 (stroke). Viral myocarditis is another recognized cause; pediatric myocarditis with a DCM phenotype is *"characterized by early-onset heart failure, significant enrichment of likely pathogenic/pathogenic variants, and poor outcome"* PMID: 34213952. Investigator-reported HFrEF etiologies span ischemic (57%), idiopathic, hypertensive, valvular, alcoholic, viral, and Chagasic.

Finding 14 — Cardiac fibrosis via inflammation → myofibroblast activation is a core downstream mechanism

HF remodeling involves injury-mediated immune cell infiltration and myofibroblast activation. *"An inflammatory to fibrotic transition corresponds with macrophage transition (M1-M2) associated with increased transforming growth factor (TGF)- β response,"* driving *"degradation-resistant deposition of extracellular proteins, especially fibrillar Collagen -I, -III and -V, and non-fibrillar Collagen-IV by active myofibroblasts"* PMID: 39805714. In angiotensin II-infused mouse hearts, platelet inhibition with clopidogrel *"inhibited Ang II infusion-induced accumulation of α -SMA(+) myofibroblasts and cardiac fibrosis"* PMID: 23887740. Cardiorenal syndrome features bidirectional heart-kidney inflammation and fibrosis.

GO terms: GO:0030198 (extracellular matrix organization), GO:0007179 (TGF-beta receptor signaling), GO:0002544 (chronic inflammation). **CL terms:** CL:0000057 (fibroblast), CL:0000186 (myofibroblast), CL:0000235 (macrophage). **UBERON:** UBERON:0002113 (kidney, secondary organ).

Finding 15 — Plasma proteomics and multi-omics reveal etiology-specific biology

High-throughput plasma proteomics (734 proteins) in 1,212 HFrEF patients identified "*a nine-protein panel (P9: C1QA, CCL4, REN, EGLN1, COL9A1, GP1BA, ITM2A, CNPY2, NT-proBNP) that improved risk classification compared with NT-proBNP alone*" (externally validated in UK Biobank) [PMID: 42258512](#). Crucially, "*in CCC [chronic Chagas cardiomyopathy] the P9 panel underperformed NT-proBNP alone (-16%), suggesting distinct underlying disease biology*" — confirming etiology-specific molecular signatures. Mitochondrial-associated lncRNAs (LIPCAR, MALAT1, H19) and circulating microRNAs are emerging biomarkers [PMID: 42236984](#); a hallmark transcriptomic signature is reactivation of the fetal gene program (NPPA, NPPB, MYH7). **Resources:** PRIDE/ProteomeXchange (proteomics), GEO (transcriptomics).

Finding 16 — Tiered prevention: primary risk-factor control is highest-yield

Primary prevention emphasizes "*Lifestyle modifications, including dietary changes, smoking cessation, physical activity, weight management, and psychosocial support*" plus pharmacologic control of blood pressure, lipids, and glucose [PMID: 41628971](#). Life's Essential 8 and cardiovascular-kidney-metabolic (CKM) staging stratify risk: higher CKM stage conferred stepwise CVD risk up to HR 5.95 (95% CI 4.75–7.45) in a 100,727-person cohort [PMID: 40346555](#). Secondary prevention uses natriuretic-peptide and global longitudinal strain (GLS) screening of stage A/B individuals (GLS screening cost-effective at €69,543/QALY [PMID: 42025682](#)). Tertiary prevention (GDMT, devices, cardiac rehab) limits complications.

Detailed Section Coverage

1. Disease Information

HF is a clinical syndrome (see Finding 1). **Identifiers:** MONDO:0005252, ICD-10 I50, ICD-11 BD10-BD1Z, MeSH D006333. **Synonyms:** congestive heart failure (CHF), cardiac failure, cardiac insufficiency, myocardial failure. Information derives predominantly from **aggregated disease-level resources** (guidelines, registries, trials) supplemented by EHR/registry-level individual-patient data (e.g., SwedeHF, MyoVasc, EUROMACS).

2. Etiology

Causal factors: ischemic (coronary artery disease, ~57% of HFrEF), hypertensive, genetic (DCM: *TTN*, *LMNA*, etc. — Finding 3), infectious (Chagas, viral myocarditis — Finding 13), valvular, alcoholic/toxic, and cardiotoxic (cancer therapies). **Genetic risk:** monogenic causal variants plus polygenic susceptibility; ~0.7% population carriage of actionable cardiomyopathy variants [PMID: 35544052]. **Environmental risk:** age, hypertension, diabetes, obesity, smoking, air/noise/light pollution [PMID: 42095252]. **Protective factors:** healthy lifestyle (Life's Essential 8), weight loss/bariatric surgery [PMID: 42591586], guideline-directed risk-factor control. **Gene-environment interactions:** genetic substrate (e.g., subclinical *Ttn* truncation) unmasked by hemodynamic stress/pressure overload [PMID: 26504781].

3. Phenotypes

See Finding 6. Phenotype types span **symptoms** (dyspnea, fatigue, orthopnea, PND), **clinical signs** (peripheral edema, elevated JVP, rales, S3 gallop), and **laboratory abnormalities** (elevated natriuretic peptides, troponin). Adult/late-onset, progressive with episodic decompensation, variable severity (NYHA I–IV). Quality-of-life impact is severe and central to disease burden (KCCQ, MLHFQ). HPO terms enumerated in Finding 6.

4. Genetic/Molecular Information

Causal genes (Finding 3): *TTN* (18%), *LMNA* (8%), *MYH7*, *MYBPC3*, *TNNT2*, *DSP*, *FLNC*, *BAG3*, *SCN5A*, *RBM20*, *DMD*. **Variant types:** *TTN* truncating variants (most common); *LMNA/FLNC* — arrhythmogenic, high-risk. **Classification:** ACMG/AMP P/LP across 45 genes [PMID: 37795486]. **Inheritance:** predominantly autosomal dominant. **Modifiers/genotype-phenotype:** LGE patterns are gene-specific (subepicardial in *DMD/DSP/FLNC*; absent/rare in *TNNT2/RBM20/*

MYH7) [PMID: 37562008](#). **Origin:** germline for inherited cardiomyopathy. **Functional consequence:** loss of function (*TTN* haploinsufficiency), structural/nuclear-envelope disruption (*LMNA*).

5. Environmental Information

Environmental factors: air pollution (particulate matter), noise, heat, chemical/light pollution [PMID: 42095252]; trace-element/metallomic dysregulation (iron, zinc, selenium, magnesium deficiency) [PMID: 42606681]. **Lifestyle:** smoking, poor diet, physical inactivity, alcohol, obesity. **Infectious agents:** *T. cruzi* (Chagas), coxsackievirus B, parvovirus B19, HHV-6, SARS-CoV-2 (myocarditis).

6. Mechanism / Pathophysiology

See Findings 5 and 14 and the Mechanistic Model below. **Pathways:** RAAS, sympathetic/ β -adrenergic signaling, TGF- β , AMPK/TFEB-lysosomal, SGLT2-mediated metabolic substrate shift. **Cellular processes:** mitochondrial dysfunction, oxidative stress, apoptosis, autophagy, inflammation, fibrosis. **Metabolic:** substrate shift from fatty acids to glucose/ketones, impaired oxidative phosphorylation. **Immune:** macrophage M1→M2 transition, β 1-AAB autoimmunity. **Molecular profiling:** fetal gene reactivation (*NPPA/NPPB/MYH7*), plasma proteomic signatures, mito-lncRNAs.

7. Anatomical Structures Affected

Primary organ: heart (UBERON:0000948), specifically myocardium (UBERON:0002349), left ventricle (UBERON:0002084). **Secondary:** lungs (pulmonary congestion, UBERON:0002048), kidneys (cardiorenal, UBERON:0002113), liver (congestion), systemic venous circulation. **Body systems:** cardiovascular (primary), renal, respiratory. **Cell types:** cardiomyocytes (CL:0000746), cardiac fibroblasts/myofibroblasts (CL:0000057/CL:0000186), macrophages (CL:0000235), microvascular endothelial cells. **Subcellular:** mitochondria (GO:0005739), sarcoplasmic reticulum, lysosome, nucleus. **Lateralization:** left-sided, right-sided, or biventricular.

8. Temporal Development

Onset: predominantly adult/geriatric; genetic DCM can present in childhood (pediatric myocarditis-DCM, median 1.4 y [PMID: 34213952]). **Onset pattern:** chronic/insidious (most) or acute (acute decompensated HF, post-MI). **Stages:** A → B → C → D (Universal Definition). **Course:** progressive with episodic decompensations; partial reversibility possible (HFimpEF, LVAD reverse remodeling). **Duration:** chronic, lifelong.

9. Inheritance and Population

Epidemiology: >64 million worldwide [PMID: 42606681]; ~1-in-4 lifetime risk [PMID: 41670570]. **Inheritance (genetic subset):** autosomal dominant DCM; overall multifactorial/polygenic. **Penetrance:** incomplete, age-dependent. **Expressivity:** variable (genotype-specific LGE, arrhythmia risk). **Demographics:** prevalence rises with age; HFrEF male-predominant; regional variation (Chagas endemic in Latin America/US).

10. Diagnostics

Laboratory/biomarkers: BNP/NT-proBNP (cornerstone), hs-troponin, galectin-3, sST2, GDF-15, MR-proANP, copeptin (Finding 7). **Imaging:** echocardiography (LVEF, GLS, diastolic function), cardiac MRI with late gadolinium enhancement (genotype-specific patterns [PMID: 37562008]). **Functional:** cardiopulmonary exercise testing, 6MWT. **Electrophysiology:** ECG (QRS duration/morphology for CRT eligibility). **Biopsy:** endomyocardial biopsy for myocarditis. **Genetic testing:** multi-gene cardiomyopathy panels (24.6–46% yield), WES/WGS; recommended for DCM/familial disease. **Clinical criteria:** 2021 Universal Definition; NYHA class. **Differential:** COPD, pulmonary embolism, valvular disease, aortic stenosis.

11. Outcome/Prognosis

Mortality: five-year rates rival solid-organ malignancies [PMID: 42606681]; HFrEF all-cause death 42.7% [PMID: 38493451]. **Etiology-specific:** Chagas worst (CV death HR 1.86 vs ischemic). **Prognostic factors:** EF class, renal function, natriuretic peptides, sex, sinus rhythm. **Complications:** recurrent hospitalization, arrhythmia, cardiogenic shock, cardiorenal failure, stroke, thromboembolism. **QoL measures:** KCCQ, MLHFQ, EQ-5D, PROMIS. **Risk models:** LIFE-Preserved (HFpEF, C-statistic 0.66–0.71 [PMID: 41810940](#)), P9 proteomic panel.

12. Treatment

See Findings 4 and 10. **Pharmacotherapy:** ARNI, beta-blockers, MRA, SGLT2i (four pillars); + vericiguat, digitoxin, ferric carboxymaltose, loop diuretics, finerenone. **Devices:** CRT, ICD. **Advanced:** LVAD (bridge/destination), heart transplantation. **Supportive/palliative:** dobutamine, opioids for end-stage symptom control. **Rehabilitation:** supervised cardiac rehab. **Personalized:** phenotype-guided GDMT sequencing, genotype-informed risk stratification.

13. Prevention

Tiered (Finding 16): **primary** (risk-factor/lifestyle control, Life's Essential 8), **secondary** (NP/GLS screening of stage A/B), **tertiary** (GDMT, rehab, comorbidity management). No vaccine except addressing infectious triggers (vector control for Chagas). Genetic counseling and cascade screening for familial cardiomyopathy.

14. Other Species / Natural Disease

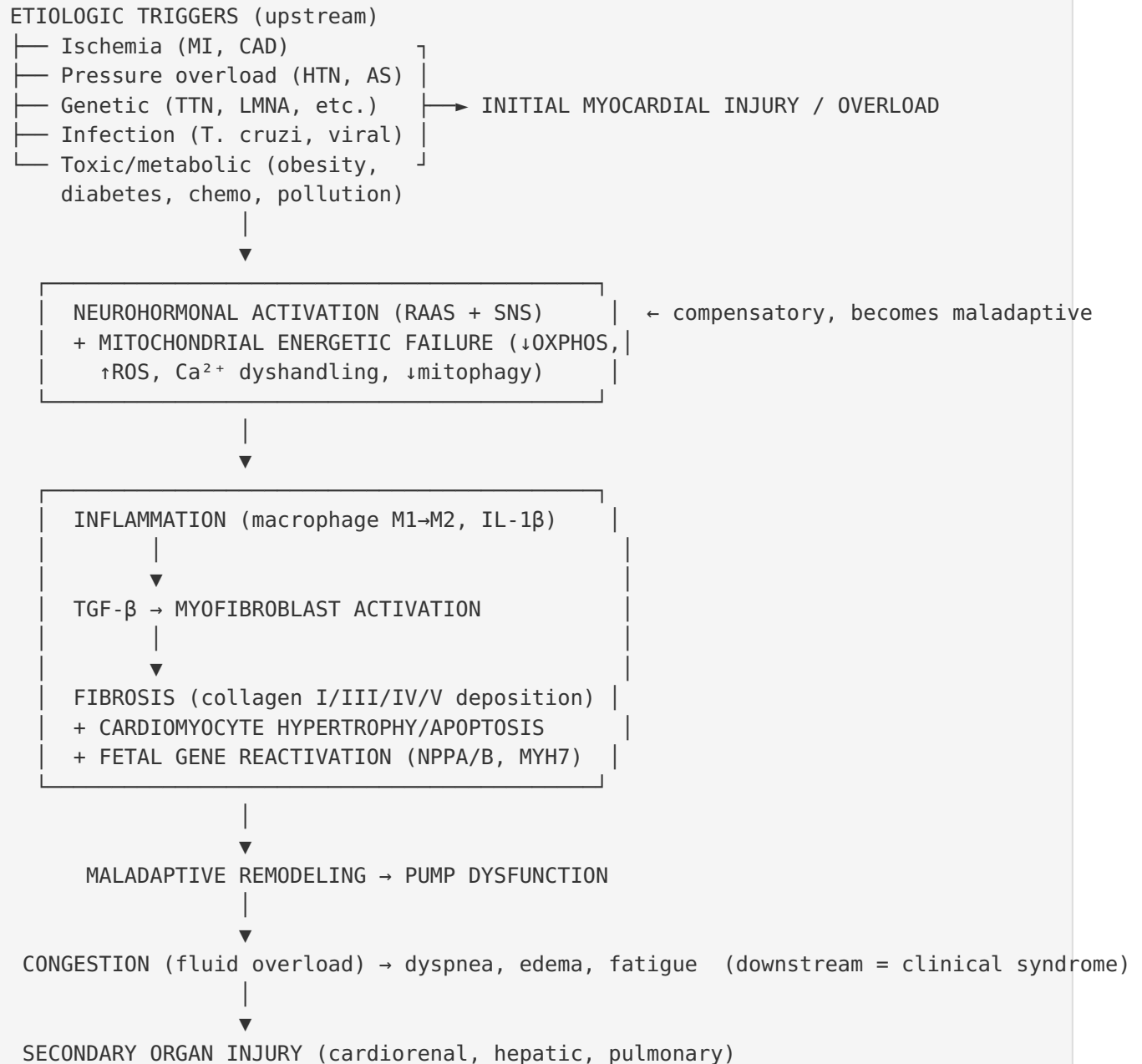
See Finding 12. Naturally occurring DCM/HF in dogs (Doberman, Boxer, Great Dane) and feline HCM (Maine Coon, Ragdoll). Strong evolutionary conservation of mitochondrial dysfunction, calcium-handling defects, and β 1-AAB autoimmunity. High veterinary relevance; canine DCM is a translational model. **Resource:** OMIA. No zoonotic transmission of HF itself (though *T. cruzi* is zoonotic).

15. Model Organisms

See Finding 11. **Mammalian:** TAC/post-MI mouse and rat, *Ttn/Lmna/Mc1r* genetic mice [PMID: 39921516](#). **Non-mammalian:** zebrafish HFpEF. **In vitro:** H9c2, neonatal cardiomyocytes, human iPSC-cardiomyocytes, organoids. **Types:** knockout, knock-in, conditional, humanized. **Limitations:** rodent heart rate/electrophysiology differ from human; models rarely capture multimorbid elderly HFpEF. **Resources:** MGI, RGD, ZFIN, IMPC, IMSR.

Mechanistic Model / Interpretation

Heart failure is best understood as a **final common pathway** onto which heterogeneous etiologies converge:



Upstream mechanisms (neurohormonal activation, mitochondrial failure) are the earliest and most therapeutically tractable — this is precisely why the four GDMT pillars work: ARNI and MRA interrupt RAAS; beta-blockers interrupt SNS; SGLT2i restore mitochondrial substrate efficiency. **Downstream mechanisms** (established fibrosis, remodeling) are harder to reverse but partially tractable (LVAD-induced reverse remodeling, HFimpEF). The **etiology-specific molecular signatures** (e.g., Chagas biology diverging from the P9 proteomic panel) demonstrate that although the syndrome is shared, its molecular substrate is not uniform — the future of HF care is phenotype- and genotype-informed precision medicine.

Evidence Base

PMID	Title (abbrev.)	Supports Finding	Evidence type
33663906	Universal Definition & Classification of HF	F1 (definition, LVEF classes)	Guideline/consensus
38493451	Registry per universal definition	F1 (subtype distribution)	Human registry
41670570	HF Prevention: Evidence Generation	F2 (lifetime risk, pre-HF)	Review
40792443	DCM genetics in Poland	F3 (TTN/LMNA, LMNA risk)	Human cohort
37795486	DCM panel yield (n=2088)	F3 (24.6% yield, 45 genes)	Human cohort
40155570	DCM genes to treatments	F3 (TTN/LMNA/MYH7)	Review
42016211	GDMT optimization in HFrEF	F4 (four pillars)	Review
41913872	SGLT2i in HF without diabetes	F4 (mechanism, trials)	Review
41052644	SGLT2i+finerenone HFmrEF/HFpEF	F4 (HR 0.69)	Cross-trial analysis
40801414	Fluid restriction in HF	F5, F6 (neurohormonal)	Review
42209899	Mitochondrial crosstalk in HF	F5 (mitochondrial dysfunction)	Review
42411500	Post-MI rat mitochondria	F5 (ROS, ↓ MMP)	Model organism
39450907	HF symptom clusters & QoL	F6 (five clusters)	Human cross-sectional
42336497	Biomarkers in HF	F7 (BNP/NT-proBNP)	Review
41166576	NP screening in diabetes	F7 (HR 4.48)	Human cohort (n=116k)
42446811	Age & HF therapy (MyoVasc)	F8 (risk factors)	Human cohort
42591586	Bariatric surgery & CV risk	F8 (obesity)	Review
42095252	Environmental risk factors	F8 (pollution)	ESC consensus
42606681	Metallomic dysregulation & CVD	F9 (64M, iron deficiency)	Systematic review

PMID	Title (abbrev.)	Supports Finding	Evidence type
40387335	LVAD myocardial repair	F10 (reverse remodeling)	Review
33464950	Cardiac microvascular EC in pressure overload	F11 (TAC model)	Model organism
26504781	TAC in Titin-truncated mouse	F11 (Ttn model)	Model organism
36536484	Zebrafish/mouse ion imbalance	F11 (zebrafish HFpEF)	Model organism
37505469	Canine DCM myosin SRX	F12 (naturally occurring)	Comparative
1338376	Doberman mitochondrial defect	F12 (conserved mechanism)	Comparative
32212256	BC 007 aptamer / β 1-AAB	F12 (autoimmune driver)	Comparative/therapeutic
41847775	Comparative prognosis of Chagas	F13 (worst prognosis)	Human pooled trials
34213952	DCM variants in pediatric myocarditis	F13 (viral/genetic)	Human cohort
39805714	Fibrosis in cardiorenal syndrome	F14 (TGF- β , M1 $\rightarrow$ M2)	Review
23887740	Clopidogrel & cardiac fibrosis	F14 (platelet/inflammation)	Model organism
42258512	Plasma proteomics in HF	F15 (P9 panel, Chagas biology)	Human + validation
41628971	Saudi CVD prevention guidelines	F16 (primary prevention)	Guideline
40346555	Life's Essential 8 / CKM stages	F16 (HR 5.95)	Human cohort (n=100k)
42025682	GLS screening cost-effectiveness	F16 (secondary prevention)	Cost-effectiveness

Limitations and Knowledge Gaps

1. **No primary dataset was analyzed.** This report is a literature/knowledge synthesis; no patient-level or omics data files were provided, so all effect sizes are quoted from published studies rather than independently re-derived.
 2. **HFpEF mechanism remains incompletely defined.** Although SGLT2i benefit HFpEF, the molecular heterogeneity of HFpEF (metabolic, inflammatory, microvascular endotypes) is not fully resolved, and models incompletely recapitulate elderly multimorbid HFpEF.
 3. **Genetic yield is partial.** Even comprehensive DCM panels identify a molecular diagnosis in only ~25–46%, leaving a large "genetically elusive" fraction and many variants of uncertain significance (VUS).
 4. **Etiology-specific biology is understudied.** The Chagas proteomic divergence signals that pooled HF trials may mask etiology-specific responses; most GDMT evidence derives from predominantly ischemic/idiopathic cohorts.
 5. **Ontology mappings are suggested, not curator-verified.** HPO/GO/CL/UBERON/NCIT terms should be confirmed against current ontology releases before database ingestion.
 6. **Epidemiology figures are global aggregates.** Regional incidence/prevalence, sex ratios, and variant geography require registry-specific validation.
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Proposed Follow-up Experiments / Actions

1. **Etiology-stratified multi-omics.** Integrate proteomic (P9-type panels), transcriptomic (fetal gene program), and metabolomic signatures stratified by etiology (ischemic vs. Chagas vs. genetic DCM) to build etiology-specific risk models — directly extending [PMID: 42258512].
2. **Prospective validation of GLS + NP screening** in stage A/B populations to quantify HF-prevention yield and cost-effectiveness beyond the German HERZCHECK simulation [PMID: 42025682].
3. **Functional characterization of VUS** in *TTN/LMNA/FLNC* using iPSC-cardiomyocytes and CRISPR editing to reclassify uncertain variants and expand diagnostic yield.
4. **HFpEF endotyping trials** pairing SGLT2i/finerenone response with mechanistic biomarkers (mitochondrial function, inflammation, microvascular density) to define treatment-responsive subgroups.

5. **Comparative-medicine studies** leveraging naturally occurring canine DCM (β 1-AAB, mitochondrial ATP defect) to test neutralizing/aptamer therapies (e.g., BC 007) as translational bridges [PMID: 32212256].
6. **Implementation science** to close the GDMT gap: real-world data show persistent under-titration (especially in the elderly [PMID: 42446811]); structured "four drugs in 4 weeks" protocols warrant pragmatic trial evaluation.
7. **Curator handoff:** verify all suggested ontology terms (MONDO, HPO, GO, CL, UBERON, NCIT, CHEBI) against current releases and attach the exact abstract snippets provided herein as evidence items.

Report compiled across 5 iterations, 16 confirmed findings, and 67 reviewed papers. Evidence types span human clinical (guidelines, registries, cohorts, pooled trials), model organism (mouse, rat, zebrafish), comparative (canine/feline), and computational (risk models, cost-effectiveness simulations).
